

**Time:** 5 minutes**Outcome:** Apply open-market operations, administered rates, and reserve tools.**Difficulty:**

Intermediate

Monetary-Policy Tools



THE CORE IDEA

The Federal Reserve influences bank reserves, the money supply, and short-term interest rates through several policy tools. Open-market purchases add reserves to the banking system, while open-market sales remove them. Administered rates change the incentives banks face when borrowing or holding reserves. Changes in reserves can affect bank lending and deposit creation, although the actual expansion of money may be smaller than the simple textbook multiplier predicts.



HOW TO RECOGNIZE IT

- An open-market purchase adds reserves; in the limited-reserves model this tends to lower short-term rates.
- An open-market sale removes reserves; in the limited-reserves model this tends to raise short-term rates.
- Lower administered rates ease monetary conditions; higher administered rates restrain them.
- In an ample-reserves system, administered rates are the main tool for controlling short-term rates.



WATCH OUT

Do not assume every tool in a policy package pushes bank lending in the same direction. The open-market purchase is expansionary, while a higher return on reserves can weaken the lending response.



WORKED EXAMPLE: OPEN-MARKET PURCHASE AND DEPOSIT EXPANSION

The central bank buys \$50 million of securities, adding \$50 million to bank reserves. With a 20% required reserve ratio, no excess reserves, and full redepositing, the simple deposit multiplier is $1 / 0.20 = 5$. Maximum deposit expansion is \$50 million $\times 5 = \$250$ million. Currency drain or additional reserve holding makes the result smaller; it cannot be calculated by multiplying 5 by the lending share.



CHECK YOURSELF

If the central bank lowers the rate it charges banks for direct loans, what is the immediate textbook effect?

**READY?**

Return to the game and master this concept.